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REVIEW

Influence of static stretching on hamstring flexibility in healthy young adults: Systematic review and meta-analysis

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ABSTRACT

The aim of the current study was to investigate the influence of static stretching on hamstring flexibility in healthy young adults by means of systematic review and meta-analysis. The search strategy included MEDLINE, PEDro, Cochrane CENTRAL, EMBASE, LILACS, and manual search from inception to June 2015. Randomized and controlled clinical trials studies that have compared static stretching to control group, and evaluated range of motion (ROM), were included. On the other hand, studies that have worked with special population such as children, elderly people, athletes, and people with any dysfunction/disease were excluded, as well as articles that have used contralateral leg as control group or have not performed static stretching. The meta-analysis was divided according to three types of tests. Nineteen studies were included out of the 813 articles identified. In all tests, the results favored static stretching compared to control group: passive straight leg raise (12.04; 95% CI: 9.61 to 14.47), passive knee extension test (8.58; 95% CI: 6.31 to 10.84), and active knee extension test (8.35; 95% CI: 5.15 to 11.55). In conclusion, static stretching was effective in increasing hamstring flexibility in healthy young adults.

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Hamstring; range of motion; systematic review

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Introduction

Hamstring muscles are well known for their great tendency to shortening (Kerkhoffs, Wieldraaijer, Tsierevelt, and Dijk, 2013; Safran, Seaber, and Garrett, 1989), which is due to their multijoint condition, their tonic postural character, and considerable amount of tensional forces to which they are constantly submitted (Davis et al., 2005). The literature shows that lack of hamstring flexibility may result in major muscle imbalances, predisposing athletes to muscle injuries (Witvrouw et al., 2003), patellar tendinopathy, and patellofemoral pain (Harvard and Ronald, 2007; Jonhagen, Nemeth, and Eriksson, 1994; Witvrouw et al., 2000), as well as facilitating the development of low back pain (Alston et al., 1966; Biering-Sorensen, 1984; Jonhagen, Nemeth, and Eriksson, 1994; Witvrouw et al., 2000). Thereby, hamstring flexibility becomes an important matter in the context of physiotherapy.

Stretching has been a widespread technique to improve flexibility. There are many stretching techniques in which this goal may be achieved such as ballistic stretching, dynamic stretching, proprioceptive neuromuscular facilitation (PNF), and static stretching (Jaggers, Swank, Frost, and Lee, 2008; Manoel, Harris-Love, Danoff, and Miller, 2008). Static stretching has been the most widely used strategy since it is relatively easy to perform, it does not require either too much time or effort, it has a low risk of injuries, and it has shown positive results in improving flexibility (Aquino et al., 2010; Ayala, Sainz de Baranda, De Ste Croix, and Santonja, 2013; Cipriani, Abel, and Pirrwitz, 2003; Ford and McChesney, 2007; Marshall, Cashman, and Cheema, 2011; Rancour, Holmes, and Cipriani, 2009).

Nevertheless, there is no consensus in the literature regarding the optimal parameters of static stretching that should be applied in order to increase hamstring flexibility. Studies present different number of sets (Aquino et al., 2010; Covert, Alexander, Petronis, and Davis, 2010), duration of stretching (de Weijer, Gorniak, and Shamus, 2003; Halbertsma and Göeken, 1994; Wiemann and Hahn, 1997), length of training (Bandy, Irion, and Briggler, 1998; Cipriani, Abel, and Pirrwitz, 2003; Marshall, Cashman, and Cheema, 2011), and ways to evaluate flexibility (Bandy and Irion, 1994; Davis et al., 2005; Waseem, Nuhmani, Ram, and Ahmad, 2009). These inconsistencies diminish the effectiveness of stretching and make it difficult to establish optimal static stretching training programs.

Hence, the aim of the current study was to investigate the influence of static stretching on hamstring flexibility

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in healthy young adults by means of systematic review and meta-analysis.

Methods

Data sources and searches

We searched the following electronic databases (from inception to June 2015): MEDLINE (accessed by PubMed), Physiotherapy Evidence Database (PEDro), The Cochrane Central Register of Controlled Trials (Cochrane CENTRAL), Centro Latino-Americano e do Caribe de Informação em Ciências da Saúde (LILACS), and EMBASE. In addition, we searched the references of published studies. The search was performed in June 2015 and comprised the following terms: "Muscle Stretching Exercises," "Exercise, Muscle Stretching," "Exercises, Muscle Stretching," "Muscle Stretching Exercise," "Range of Motion, Articular," "Joint Range of Motion," "Joint Flexibility," "Flexibility, Joint," "Range of Motion," "Passive Range of Motion," combined with a high sensitivity combination of words used in the search for randomized clinical trials (Robinson and Dickersin, 2002). We included publications in English, Spanish, and Portuguese. The complete search strategy used for the MEDLINE database is shown in Table 1 (Moher, Liberati, Tetzlaff, and Altman, 2009). Details for the other strategies used are available on request.

Table 1. Complete search strategy used for MEDLINE.

#1	(randomized controlled trial[pt] OR controlled clinical trial[pt] OR randomized controlled trials[mh] OR random allocation[mh] OR double-blind method[mh] OR single-blind method[mh] OR clinical trial[pt] OR clinical trials[mh] OR ("clinical trial"[tw]) OR ((singl*[tw] OR doubl*[tw] OR trebl*[tw] OR tripl*[tw]) AND (mask*[tw] OR blind*[tw])) OR ("latin square"[tw]) OR placebos[mh] OR placebo*[tw] OR random*[tw] OR research design[mh:noexp] OR follow-up studies [mh] OR prospective studies[mh] OR cross-over studies[mh] OR control*[tw] OR prospectiv*[tw] OR volunteer*[tw]) NOT (animal[mh] NOT human[mh])
#2	"Muscle Stretching Exercises"[Mesh] OR "Exercise, Muscle Stretching" OR "Exercises, Muscle Stretching" OR "Muscle Stretching Exercise" OR "Dynamic Stretching" OR "Stretching, Dynamic" OR "Isometric Stretching" OR "Stretching, Isometric" OR "Active Stretching" "Stretching, Active" OR "Static-Active Stretching" OR "Static Active Stretching" OR "Stretching, Static-Active" OR "Static Stretching" OR "Stretching, Static" OR "Passive Stretching" OR "Stretching, Passive" OR "Relaxed Stretching" OR "Stretching, Relaxed" OR "Static-Passive Stretching" OR "Static Passive Stretching" OR "Stretching, Static-Passive" OR "Ballistic Stretching" OR "Stretching, Ballistic" OR "Proprioceptive Neuromuscular Facilitation (PNF) Stretching"
#3	"Range of Motion, Articular"[Mesh] OR "Joint Range of Motion" OR "Joint Flexibility" OR "Flexibility, Joint" OR "Range of Motion" OR "Passive Range of Motion" OR "muscle length
#4	#1 AND #2 AND #3

Studies selection

We included randomized clinical trials and controlled clinical trials that assessed the influence of static stretching (passive or active) compared to a control group (no intervention) and evaluated hamstring flexibility in healthy young adults. The following exclusion criteria were used: (1) samples comprised of people with any disease/dysfunction; (2) samples formed by athletes; (3) samples of people under 18 or over 40 years old; (4) non-application of static stretching; (5) no assessment of range of motion (ROM); and (6) contralateral leg used as control group. Two investigators, in duplicate, independently evaluated titles and abstracts of all articles identified by the search strategy. All abstracts that did not provide sufficient information regarding the inclusion and exclusion criteria were selected for full-text evaluation. In the second phase, the same reviewers independently evaluated the full-text articles and made their selection in accordance with the eligibility criteria. Disagreements between reviewers were solved by consensus or through a third person review.

Data extraction

Using standardized forms, the same two reviewers independently conducted data extraction with regard to the methodological characteristics of the studies, interventions, participants, and outcomes; disagreements were also solved by consensus. The main outcome extracted was ROM.

Quality assessment

Study quality assessment included adequate sequence generation, allocation concealment, blinding of outcome assessors, use of intention-to-treat analysis, and description of losses and exclusions. Use of intention-to-treat analysis was considered as a confirmation on study assessment that the number of participants randomized and the number analyzed were identical, except for patients lost to follow-up or who withdrew consent for study participation. Studies without a clear description of these characteristics were considered as unclear or not reporting the latter. The same two reviewers independently performed quality assessment.

Data synthesis and analysis

Pooled-effect estimates were obtained by post-intervention values. Calculations were performed using a random-effects method. A p-value ≤ 0.05 and confidence interval of 95% (95% CI) were considered statistically significant. Statistical heterogeneity of the treatment effects among

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studies was assessed using Cochran's Q test and the inconsistency I^2 test, in which values above 25% and 50% were considered indicative of moderate and high heterogeneity, respectively (Higgins, Thompson, Deeks, and Altman, 2003). All analyses were conducted using Review Manager version 5.2. Because of the different techniques used to evaluate ROM, it was necessary to separate the studies for the meta-analysis according to three types of test: (1) passive straight leg raise; (2) passive knee extension (full knee extension as 0°); and (3) active knee extension (90° knee flexion as 0°). We explored heterogeneity between studies by re-running the meta-analyses removing one paper at a time to check whether some individual study explained heterogeneity. Sensitivity analysis could not be performed since we were unable to find subgroups of studies that could provide relevant clinical information within the analyses performed.

Results

Description of studies

The search strategy yielded 813 articles, among which 58 studies were considered as potentially relevant and retrieved for detailed analysis. Nineteen of these studies met the eligibility criteria and were included in the systematic review (n = 689). Figure 1 shows the flow diagram of the studies included in this review and Table 2 summarizes the characteristics of these studies.

Risk of bias

Of the studies included in the systematic review, 26.3% presented an adequate sequence generation, none presented reported allocation concealment; 42.1% had blinded assessment of outcomes; 36.8% described losses to follow-up and exclusions; and 63.1% of the studies used the intention-to-treat principle for statistical analyses (Table 3).

Effects of interventions

Range of motion

The quantitative analysis of the ROM was carried out grouping the articles that have used the same ROM assessment technique, as described above.

Passive straight leg raise test

Eight studies (Ayala and de Baranda Andújar, 2010; Ayala et al., 2013; Cipriani et al., 2012; Gajdosik, 1991; Gribble, Guskiewicz, Prentice, and Shields, 1999; Halbertsma and Göeken, 1994; Marshall, Cashman, and Cheema, 2011; Wiemann and Hahn, 1997) (n = 291) evaluated the influence of the static stretching on the ROM, using as evaluation method the passive straight leg raise test. We observed that static stretching provides higher gain in ROM compared to the control group (12.04°; 95% CI: 9.61 to 14.47; I^2: 2%) (Figure 2).

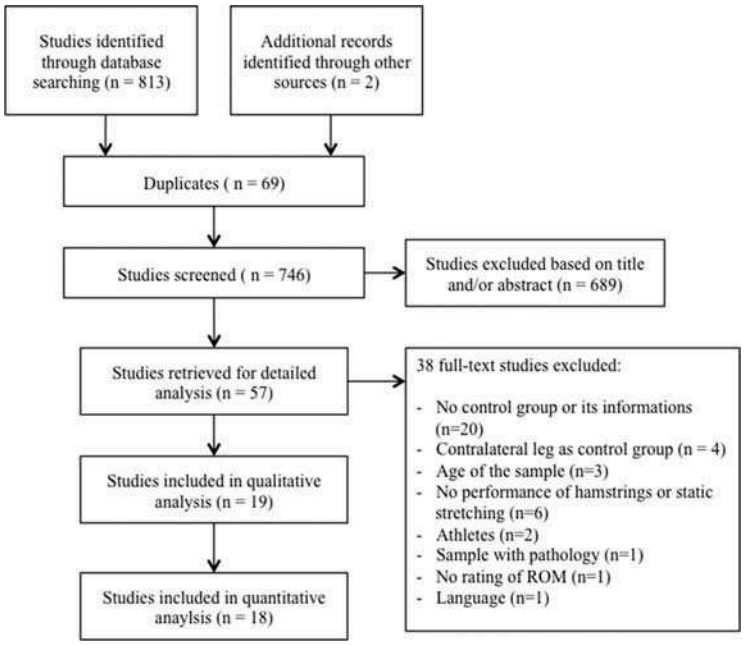


Figure 1. Flowchart of studies included in the review.

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Table 2. Characteristics of included studies.

Study	Subjects/mean age ± SD	Technique	Outcome	Duration	Protocol
Ayala and de Baranda Andújar (2010)	150/21.3 ± 2.5	Static stretching active	PSLR	6 × 30 s	12 weeks, 3/weeks
Ayala et al. (2013)	138/21.9 ± 1.7	Static stretching active	PSLR	6 × 30 s—rest 20 s	12 weeks, 3/weeks
Bandy and Irion (1994)	57/26.5 ± 5.3	Static stretching	PKE	1 × 30 s	6 weeks, daily
Bandy et al. (1997)	93/26.2 ± 5.1	Static stretching	PKE	1 × 30 s	6 weeks, daily
Bandy et al. (1998)	58/26.2 ± 5.7	Static stretching	PKE	1 × 30 s	6 weeks, daily
Cipriani et al. (2012)	53/24 ± 5.5	Static stretching	PSLR	2 × 30 s—rest 10 s	4 weeks, 3/weeks
Covert et al. (2010)	32/21.6 ± 2.7	Static stretching	PKE	1 × 30 s	4 weeks, 3/weeks
Ford and McChesney (2007)	32/22.1 ± 3.0	Static stretching	AKE	5 × 30 s—rest 10 s	Acute effect
Gajdosik (1991)	24/25.3 ± 5.6	Static stretching	AKE and SLR	10 × 15 s—rest 15 s	3 weeks, daily
Gribble et al. (1999)	42/19.67 ± 1.55	Static stretching	AKE and PSLR	4 × 30 s—rest 30 s	6 weeks, 3/weeks
Halbertsma and Göeken (1994)	14/27.2 ± 5.3	Passive stretching	PSLR	10 min (twice a day)	4 weeks, daily
Li et al. (1996)	39/29.3 ± 3.5	Static stretching	AKE and PSLR	10 × 15 s—rest 15 s	3 weeks, daily
Lim et al. (2014)	48/22.25 ± 2.29	Static stretching	AKE	1 × 30 s	Acute effect
Marshall et al. (2011)	22/22.7 ± 3.8	Passive stretching	PSLR	3 × 30s—4 stretchings	4 weeks, 5/weeks
Milazotto et al. (2009)	25/21 ± 4	Static stretching	AKE	10 × 30 s	6 weeks, 3/weeks
O'Hora et al. (2011)	45/25 ± 2.8	Static stretching	PKE	1 × 30 s	acute effect
Rancour, Terry, Holmes and Cipriani (2010)	20/Not informed	Static stretching	PSLR	1 × 30 s (twice a day)	4 weeks, daily
de Weijer et al. (2003)	56/25.2 ± 3.1	Static stretching	AKE	3 × 30—rest 10 s	Acute effect
Wiemann and Hahn (1997)	69/Not informed	Static stretching	PSLR	3 × 15—3 stretchings	Acute effect

PSLR, Passive straight leg raise; PKE, Passive knee extension; AKE, Active knee extension.

Table 3. Risk of bias of included studies.

Study	Adequate sequence generation	Allocation concealment	Blinding of outcomes assessors	Description of losses and exclusions	Intention-to-treat analysis
Ayala and de Baranda Andújar (2010)	Not informed	Not informed	Yes	Not informed	Yes
Ayala et al. (2013)	Yes	Not informed	Yes	Yes	No
Bandy and Irion (1994)	Not informed	Not informed	Yes	Yes	No
Bandy et al. (1997)	Not informed	Not informed	Not informed	Yes	No
Bandy et al. (1998)	Not informed	Not informed	Yes	Not informed	Yes
Cipriani et al. (2012)	Not informed	Not informed	Yes	Yes	No
Covert et al. (2010)	Yes	Not informed	Yes	Yes	No
Ford and McChesney (2007)	Not informed	Not informed	Not informed	Not informed	Yes
Gajdosik (1991)	Not informed	Not informed	Not informed	Not informed	Yes
Gribble et al. (1999)	Not informed	Not informed	Not informed	Yes	Yes
Halbertsma and Göeken (1994)	Not informed	Not informed	Not informed	Yes	No
Li et al. (1996)	Not informed	Not informed	Not informed	Not informed	Yes
Lim et al. (2014)	Yes	No	Not informed	Not informed	Yes
Marshall et al. (2011)	Not informed	No	Not informed	Not informed	Yes
Milazotto et al. (2009)	Not informed	Not informed	Yes	No	No
O'Hora et al. (2011)	Yes	Not informed	Not informed	Not informed	Yes
Rancour et al. (2010)	Yes	Not informed	Not informed	Not informed	Yes
de Weijer et al. (2003)	Yes	Not informed	Yes	Not informed	Yes
Wiemann and Hahn (1997)	Not informed	Not informed	Not informed	Not informed	Yes

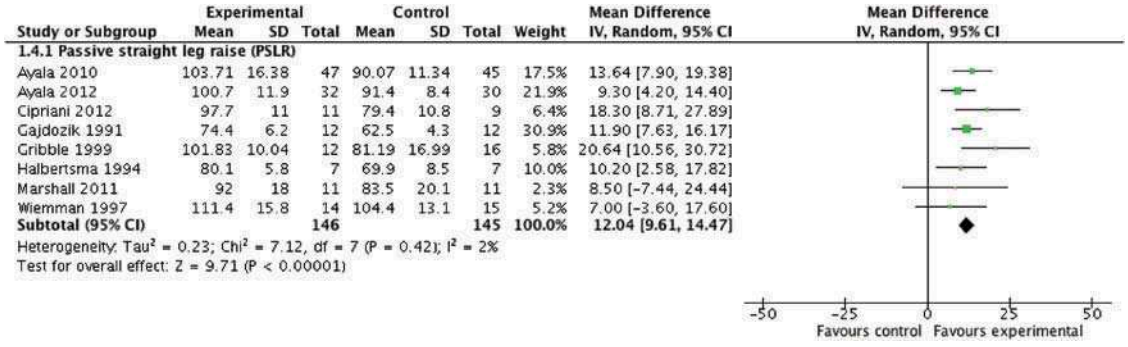


Figure 2. Analysis of passive straight leg raise test.

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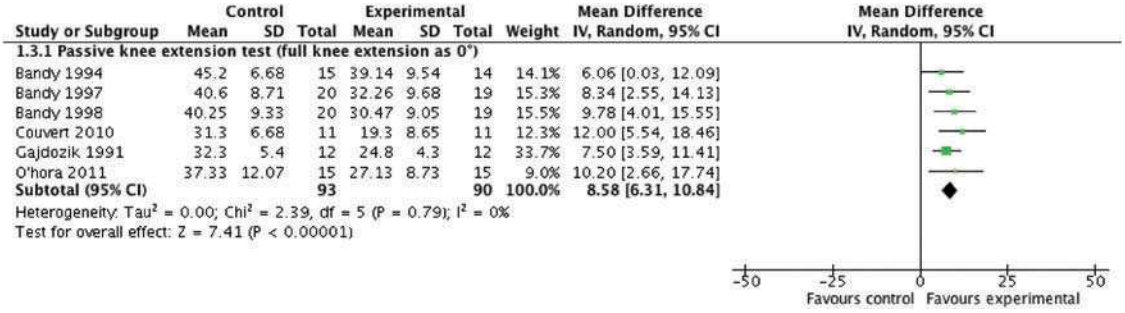


Figure 3. Analysis of passive knee extension test.

Passive knee extension test

Six studies (Bandy and Irion, 1994; Bandy, Irion, and Briggler, 1997; Bandy, Irion, and Briggler, 1998; Covert, Alexander, Petronis, and Davis, 2010; Gajdosik, 1991; O'Hora et al., 2011) (n = 183) evaluated the influence of static stretching on the ROM, using as evaluation method the passive knee extension test. Static stretching provides higher gain in ROM compared to the control group (8.58; 95% CI: 6.31 to 10.84; I^2: 0%) (Figure 3).

Active knee extension test

Five studies (de Weijer, Gorniak, and Shamus, 2003; Ford and McChesney, 2007; Gribble, Guskiewicz, Prentice, and Shields, 1999; Lim, Nam, and Jung, 2014; Millazoto, Corazzina, and Liebano, 2009) (n = 184) evaluated the influence of the static stretching on the ROM, using active knee extension test (knee flexion 90° considered as 0°). However, only four studies were suitable for meta-analysis (de Weijer, Gorniak, and Shamus, 2003; Ford and McChesney, 2007; Gribble, Guskiewicz, Prentice, and Shields, 1999; Millazoto, Corazzina, and Liebano, 2009) (n = 136). We observed that static stretching provides higher gain in ROM compared to the control group (8.35°; 95% CI: 5.15 to 11.55; I^2: 0%) (Figure 4).

Discussion

This systematic review and meta-analysis' findings elicit that regardless of the parameters used, static stretching was effective in increasing hamstring flexibility in healthy young adults compared to a control group (no stretching). However, there was no consensus in the literature about optimal static stretching parameters to enhance hamstring flexibility.

The mechanisms responsible for increasing muscle length after stretching are not completely understood (Weppler and Magnusson, 2010). The improvements in ROM observed after static stretching training might be explained by the fact that in static stretching there is a great possibility of increasing the number of sarcomeres in series (muscle length) due to longer exposure to the stresses generated in the specific degree of stretching, which remains constant (Bandy and Sanders, 2001). In addition, stretching increases viscoelasticity and decreases stiffness of muscular and connective tissues (Halbertsma, Van Bolhuis, and Göken, 1996; Magnusson, Simonsen, Aagaard, and Kjaer, 1996), which enhances muscular extensibility. Nonetheless, many studies (Ben and Harvey, 2010; Folpp, Deall, Harvey, and Gwinn, 2006; Konrad and Tilp, 2014; Law et al., 2009) have refuted the existence of muscular mechanical adaptation after static stretching. Instead,

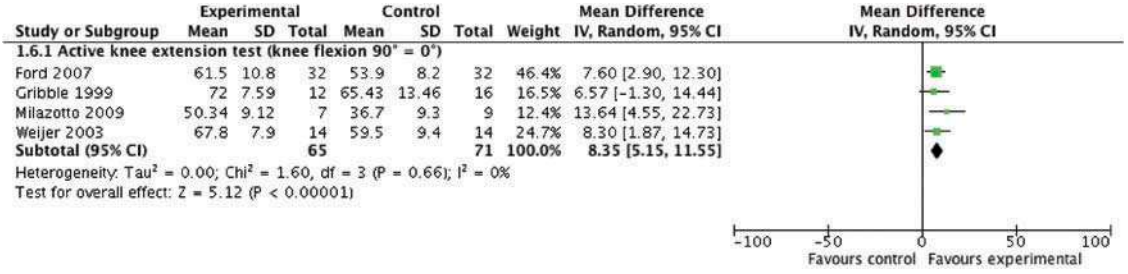


Figure 4. Analysis of active knee extension test (knee flexion 90° considered as 0°).

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they state that what appears to be a mechanical change in muscular extensibility is in fact just an increased sensory capacity of tolerating the discomfort associated with stretching of tight muscles. Regardless of the mechanisms behind flexibility improvements, the benefits of both stretching and adequate flexibility seem to be real.

Stretching itself enhances blood supply in joints and muscles, helping to warm them up, which improves functional performance during sports and activities of daily living (Savelberg and Meijer, 2003). Even though there is substantial evidence suggesting that stretching performed prior to physical activities may decrease muscle performance (Behm, Bambury, Cahill, and Power, 2004; Cramer et al., 2004; Cramer et al., 2007; Nelson, Guillory, Cornwell, and Kokkonen, 2001), we recommend a regular practice of static stretching due its satisfactory effect on flexibility as demonstrated in the present study.

As important as stretching are the flexibility improvements acquired as a result of stretching training. Adequate hamstring extensibility may prevent or relieve symptoms of both sacroiliac joint pain (Massoud Arab, Reza Nourbakhsh, and Mohammadifar, 2011) and low back pain (Alston et al., 1966; Biering-Sorensen, 1984). Likewise, proper muscular flexibility seems to decrease the risk of musculotendinous injuries (Cross and Worrell, 1999; Mahieu et al., 2006; Witvrouw et al., 2003). Flexibility also seems to have an important role in muscle function, improving results of various types of activities (Shrier, 2004). According to Ferreira, Teixeira-Salmela, and Guimaraes (2007) and LaRoche, Lussier, and Roy (2008) increased flexibility from long-term stretching training enhances muscle performance, which, in turn, improves functional capacity.

This review presents several methodological strengths such as the formulation of a focused research question, evaluation of a single stretching technique (static stretching), and a specific sample (healthy young adults). Also, a strategy for a sensitive and comprehensive search for the location of all studies in this topic was held. Furthermore, to the best of our knowledge, the current study is the first meta-analysis addressing the influence of stretching on flexibility. Making a parallel between our study and the review by Decoster, Cleland, Altiery, and Russel (2005), we are able to state that our findings seem more conclusive. This is probably related to the fact that we have focused our study on static stretching only, which enables us to carry out a meta-analysis.

One factor that negatively interfered in the data analysis was the difference between training protocols, with respect to both duration of stretching and length of training. This discrepancy did not allow us to conduct a meta-analysis compiling all studies, and it was not possible to carry out relevant sensitivity analyses that would enable comparisons between stretching parameters found in the literature. Another point that should be highlighted as a limitation is the moderate methodological quality of the included articles. None of the studies reported allocation concealment. Less than half of the studies blinded the outcome assessors or explained the generation of the random sequence. These data show that greater concern about the internal validity of the studies is required in order to obtain more reliable data.

Conclusion

This systematic review showed that static stretching was effective in increasing hamstring flexibility in healthy young adults, regardless of the stretching parameters used among the included studies. However, due to the methodological heterogeneity between the studies, it was not possible to determine optimal stretching training parameters. Future systematic reviews and studies should either analyze other stretching techniques or compare static stretching with other approaches to find out the most effective way to improve flexibility.

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Declaration of interest

The authors have no conflicts of interest and did not receive financial assistance to perform this review.

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